Referee-style Report: *Matrix Theory Reloaded: A BPS Road to Holography*

Comments on `paper.tex`

1. Principal achievement

The paper gives a unified target-space account of the BPS decoupling limits that produce matrix p-brane theories (MpT) on Dp-branes, and it uses that account to connect matrix theory to holography. Three results stand out:

- **Geometry of the limit.** The limit $\omega \to \infty$ produces a flat non-Lorentzian background with a codimension-$(p+1)$ foliation and a p-brane boost symmetry. The paper generalises this to curved backgrounds through vielbeins, and it shows that the near-horizon limit of a Dp-brane geometry is equivalent to a BPS decoupling limit applied at asymptotic infinity. This gives a concrete mechanism by which AdS/CFT appears as a decoupling limit of a single brane stack.
- **A holographic classification scheme.** Applying two successive limits yields new holographic pairs, some with non-Lorentzian bulk geometry. The paper packages these into a DLCQ^n/DLCQ^m conjecture, in which each extra DLCQ on the field-theory side corresponds to one extra near-horizon limit in the bulk. The D1-D5 system is worked through as an example of DLCQ^0/DLCQ^2.
- **Intrinsic $T\bar{T}$-type flows.** The bulk geometry is reinterpreted as a p-brane generalisation of the $T\bar{T}$ deformation acting on the non-Lorentzian field theory. This removes the need for a limiting procedure. Explicit flow equations are derived for Dp-brane DBI actions, and the $p = 1$ case reduces to standard $T\bar{T}$.

The most useful single contribution is the first: it turns a vague analogy between matrix theory and AdS/CFT into a precise statement about which decoupling limit produces which bulk geometry.

2. Principal drawback

The central claims are stronger than the evidence the paper provides. The authors say so themselves in the Outlook, but the abstract and introduction do not reflect the qualification.

- **Non-Lorentzian bulk geometries are not checked against equations of motion.** The new bulk solutions, which come from rescaling the harmonic-function length scale ℓ or from smearing, are not shown to solve the MpT limit of supergravity, including its torsional constraints. The Outlook explicitly lists this as unverified. The bulk side of the new non-Lorentzian holographic pairs therefore rests on an ansatz, not a solution.
- **The DLCQ^n/DLCQ^m conjecture is not tested beyond known cases.** The only non-trivial checks are the classic AdS/CFT examples, which the framework was built to reproduce, and the D1-D5 example. No new prediction is made and checked, so the conjecture is at present a reorganisation of known examples.

- **The $T\bar{T}$-type flows are only partly derived.** The general equation is solved for free scalars with an abelian gauge field. The $p = 3$ equation is given only to linear order in λ_p, and the non-abelian case, which is what matrix theory needs, is left open. The authors acknowledge that the endpoint for non-abelian theories is unknown.

The second drawback is the most serious for impact, because the holographic claims are the paper's headline.

3. Most worthwhile follow-up idea

Proposed project: Check whether the rescaled non-Lorentzian near-horizon geometries for the M2-brane and D2-brane cases solve the equations of motion and torsional constraints of the corresponding non-relativistic (MpT) supergravity.

- **Why this one.** The paper flags this check as open. It is a concrete, bounded calculation with a clear success criterion: the metric either satisfies the non-relativistic equations of motion or it does not. Both outcomes are informative. A positive result would make non-Lorentzian holography a real solution class. A negative result would identify which ansatz fails and sharpen the rules for constructing such duals.
- **Feasibility.** The M2-brane non-relativistic supergravity is already analysed in the literature the paper cites (Lambert et al. and Bergshoeff et al.), so there is a template to follow. The D2 case, with its gauge field, is a natural first extension. The work is a few months of computation for a graduate student.
- **Impact.** It tests the paper's most important physical claim directly. It also bears on the open questions about isometry groups and finite-temperature non-Lorentzian holography that the Outlook raises.

An alternative with higher ceiling but lower feasibility is to derive the non-abelian extension of the $p = 2$ flow equation. That would connect the $T\bar{T}$ picture to the matrix-theory side and could turn the DBI-to-matrix link into a genuine deformation statement. It is harder, because the non-abelian structure is not yet fixed, so I rank it second.